

Vivek Keerthiveetil

AI Engineer | Generative AI & Applied AI Systems

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EXPERIENCE

Gauss Robotics GmbH, Braunschweig, Germany — *Werkstudent*

December 2024 - May 2026

- Built a modular end-to-end PyTorch-based inference using Vision-Language Models (VLMs) to analyze robotic task failures videos and generate explanation and correction guidance as part of Master's thesis.
- Developed an LLM-as-judge evaluation framework to score free-form model outputs across failure analysis and correction tasks, focusing on output quality and consistency.
- Evaluated and implemented prompting strategies to analyze their effect on AI output and improve model performance.
- Developed a FastAPI and Gradio-based demo service for multimodal model inference, exposing AI workflows through API-driven interfaces and using 4-bit BitsAndBytes quantization to reduce memory usage and support lower-latency prototype testing.
- Automated robotic data acquisition by coordinating robot execution, camera control, Amazon IVS recording, and S3 storage, reducing repeated manual recording and file-management steps.
- Implemented event-driven AWS Lambda processing workflows to convert recorded video streams into inference-ready files for downstream multimodal model inference.
- Supported deployment of real-time RTSP-to-cloud video streaming pipelines using FFmpeg, Amazon IVS, Kubernetes, and Helm, while maintaining version-controlled workflows through Git and CI/CD-oriented tools.
- Developed and evaluated a Kubernetes-orchestrated proof of concept for real-time streaming between edge services and AWS EKS, using the results to support architectural decisions toward a more stable cloud-based streaming solution.

Elenia Institute for High Voltage Technology and Power Systems TU Braunschweig, Braunschweig, Germany — *HiWi*

June 2024 - November 2024

- Developed Python workflows to analyze lightning impulse voltage data, extract key event metrics, and automate statistical analysis.
- Built a lightweight Tkinter GUI that enabled non-technical users to run analysis scripts, inspect results, and store outputs reproducibly.

DXC Technology, Gurgaon, India — *Software Engineer*

April 2022 - September 2023

- Maintained API-driven data integration workflows in enterprise production environments.
- Built SQL-based logging and monitoring systems for system diagnostics.
- Collaborated with stakeholders to deliver scalable and maintainable data solutions.

Job Related Abilities

Generative AI & RAG Systems

- Experience with LLMs, RAG, and Vision-Language Models and agent workflows
- Able to evaluate model behavior, reasoning quality, and failure cases

End-to-End AI System Development

- Experience building ML workflows from data processing to deployment
- Understanding of scalable pipelines and real-world system constraints

Problem Solving & Ownership

- Structured approach to ambiguous technical problems
- Able to balance model performance, latency, and system complexity

Collaboration & Communication

- Experience working in cross-functional and international engineering teams
- Able to explain technical concepts and trade-offs clearly to technical and non-technical stakeholders

Languages

English - C1

German - B1 (actively improving)

Hindi - native

Tata Consultancy Services , Bengaluru,India — Software Engineer

May 2018 - September 2019

- Developed RESTful APIs and automated enterprise data workflows to support backend integration processes.
- Improved system efficiency through automation and process optimization.

EDUCATION

TU Braunschweig, Braunschweig, Germany — MSc Data science

October 2023 - May 2026

- Finished second in DLR project based course on semantic segmentation based on the IOU score.
- **Master's Thesis:** Failure Analysis and Correction Guidance for Robotic Manipulation Using the RoboFAC Vision-Language Model(in collaboration with Gauss Robotics).
- Grade: 1,4 (Till date)

SRM Institute of Science and Technology, Kattankulathur, India — B.Tech Computer Science and Engineering

May 2016 - May 2020

- **Bachelor Thesis:** Smart Medical shop, Project aimed at using RNN -LSTM based model to recognise medicines on prescription.
- Grade: 82.22%

PROJECTS

Agentic Job Application Workflow API

- Built a FastAPI-based LLM agent that parses job descriptions, matches candidate profiles, generates tailored application content, and reviews outputs for unsupported claims.
- Implemented manual, LLM-planned, and LangGraph workflows using modular tools, shared state, conditional routing, revision control, and traceable step history.
- Used structured JSON outputs and Pydantic validation across job analysis, profile matching, content generation, honesty checking, and revision stages.

RAG-Based Semantic Alignment System

- Built a modular RAG pipeline using MiniLM sentence embeddings and FAISS for semantic retrieval over resume documents.
- Implemented PDF parsing, document chunking, embedding generation, vector indexing, top-k retrieval, and structured LLM-based resume-job alignment analysis.
- Designed the system to generate grounded outputs with retrieved resume evidence, including alignment summaries, missing skill explanations, bullet improvements, and impact suggestions.

Object Detection & Model Evaluation

- Implemented and compared YOLO, Faster R-CNN, and DETR for PCB defect detection.
- Designed controlled experiments to evaluate robustness and architectural trade-offs.
- Evaluated models using mAP, IoU, precision, and recall.

Technical Proficiency

Programming:

Python, SQL, Git, Linux

Backend & APIs:

FastAPI, REST APIs, Pydantic

Generative AI:

LLMs, Vision-Language Models, Agent Workflows, LangGraph, RAG, Prompt Engineering, OpenAI API, Hugging Face Transformers, FAISS

ML & Deep Learning:

PyTorch, TensorFlow, CNNs, Object Detection, Semantic Segmentation, Model Evaluation

Cloud & Deployment:

AWS (S3, Lambda, IVS), Docker, Kubernetes, Helm, CI/CD concepts

Data & Streaming:

ROS/rosbag, RTSP, WebRTC, FFmpeg, Distributed Systems

References

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